

Ben Finkelshtein

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Education

University of Oxford (Expected graduation May 2026)

January 2023 - Present

Doctor of Philosophy (PhD) in Computer Science

Oxford, United Kingdom

- Titled "Efficient and Expressive Neural Networks for Learning on Graphs"
- Supervised by DeepMind Chair of AI Prof. Michael M. Bronstein and Prof. İsmail İlkan Ceylan
- Awarded the **Clarendon Scholarship of over £200,000** towards tuition fees and living costs (< **1% acceptance**)

Technion - Israel Institute of Technology

November 2019 - June 2022

Master of Science in Computer Science

Haifa, Israel

- Supervised by Prof. Alexander Bronstein and Prof. Chaim Baskin
- **GPA: 99.1; Graduated with highest honors (summa cum lauda, top 1%)**

Technion - Israel Institute of Technology

November 2013 - July 2017

Bachelor of Science in Electrical Engineering and Bachelor of Science in Physics

Haifa, Israel

- Completed two separate B.Sc as a Fellow in the IDF's competitive Psagot program (< **1% acceptance**)
- **GPA: 92.1; Graduated with great honors (magna cum lauda, top 3%)**

Selected Publications

First @ [Actions Speak Louder than Prompts: A Large-Scale Study of LLMs for Graph Inference](#) ICLR 2026 Oral
A principled analysis of LLM capabilities and limitations in processing graph information

Senior @ [Of Graphs and Tables: Zero-Shot Node Classification with Tabular Foundation Models](#) Under review
We reformulate node classification as a tabular problem and show tabular foundation models outperform graph ones

First @ [Equivariance Everywhere All At Once: A Recipe for Graph Foundation Models](#) NeurIPS 2025
We derive the first graph foundation model that is provably universal and achieves SOTA zero-shot performance

Co-First @ [Position: Graph Learning Will Lose Relevance Due To Poor Benchmarks](#) ICML 2025
We call for a paradigm shift toward meaningful benchmarks, rigorous evaluation, & more domain expert involvement

First @ [Learning on Large Graphs using Intersecting Communities](#) NeurIPS 2024
A fundamentally different pipeline for learning on very large non-sparse graphs using intersecting communities

First @ [Cooperative Graph Neural Networks](#) ICML 2024
A new graph signal processing paradigm in which each node can choose a different communication strategy

Blog Posts

- Microsoft official post [Actions Speak Louder Than Prompts: Rethinking How LLMs Reason Over Graph Data](#)
- Towards Data Science's [Graph & Geometric ML in 2024](#) and [Cooperative Graph Neural Networks](#)

Professional Experience

Google Research; Student Researcher

September 2025 – Present

- Devising the first **Graph Foundation Model for production** at Google with Bryan Perozzi and Michael Galkin
- Enabling **node-, link-, and graph-level tasks** across diverse domains, graph structures, and feature modalities.

Microsoft Research HQ; Student Researcher

June 2025 – September 2025

- Researched **LLM-based graph reasoning**, via prompting, tool-use, and code-generation paradigms on graph data
- [Actions Speak Louder than Prompts: A Large-Scale Study of LLMs for Graph Inference](#); Oral Presentation at ICLR

eBay; Applied Researcher

June 2022 – January 2023

- **Engineered LLMs** to match queries with the most relevant landing pages for search engine optimization
- Developed a keyword extraction technique, combined it with **GPT3**, and **subsequently fine-tuned the process**
- Achieved a 30% increase in landing page click-through rates (CTR) within the first three months of implementation

SKF Group - AI Center of Excellence; Data Scientist

August 2020 – May 2021

- Designed, customized, and implemented end-to-end machine learning pipelines, from ideation to production
- Led research in event prediction, **anomaly detection, and time series forecasting** to aid machinery fault diagnosis
- Designed an event-based evaluation metric that improved the company's anomaly detection from 78% to 90%

Rafael Advanced Defense Systems; Algorithm Developer

October 2017 – July 2020

- Specialized military service assignment (**1 recruit per year**)
- **Created a neural network which predicts n -body problem solutions** for classified company use
- Engineered generative models to predict molecular dynamics pathways, utilizing physical simulations
- Created new **generative models to analyze stable & transient protein folding conformations**

Awards

G-Research Grant - £2000	September 2024
Graduated Top 1% from a Master of Science in Computer Science - £500	July 2023
Clarendon Scholarship - over £200,000	July 2022
M.Sc Study Scholarship - over £45,000	December 2019
Graduated Top 3% from a Bachelors of Science in Electrical - £100	July 2018
Graduated Top 3% from a Bachelors of Science in Physics - £100	July 2018

Selected Talks

BGU University @ Israel : "A Recipe for Graph Foundation Models"	December 2025
Learning On Graphs conference @ USA : "A Recipe for Graph Foundation Models"	December 2025
Microsoft Research Headquarters @ USA : "LLM for Graph Inference"	September 2025
ISTA University @ Austria : "Learning on Large Graphs"	November 2024
Technion University @ Israel : "Learning on Large Graphs"	September 2024
RWTH University @ Germany : "Cooperative Graph Neural Networks"	July 2024
Learning On Graphs conference @ United Kingdom : "Cooperative Graph Neural Networks"	December 2023

Teaching Experience

University of Oxford; Head Teaching Assistant

January 2023 – Present

- Served as head TA for the **Geometric Deep Learning and Graph Representation Learning** courses
- Created the coding assignments for the different classes
- Guided 30 graduate students through coursework, while in charge of 100 students
- Led 10 teaching assistants, coordinating effective tutorial sessions, and timely support for students

Blue Education; Private Tutor

July 2023 – Present

Taught Math, Physics and Computer Science for A-Levels, SAT, and college exams as an online part time tutor

Technion - Israel Institute of Technology; Head Teaching Assistant

March 2021 – June 2022

- Created all of the [tutorials and code notebooks](#)
- **Conducted hybrid online and classroom lectures** and practice sessions for the Geometric Deep Learning course
- Supervised 40 graduate and undergraduate students including final course projects

Matematix - High-School and College Exam Preparation Center; Teacher

July 2012 – November 2013

- Led sessions of thirty students twice a week
- **Taught Mathematics, Physics, English and Coding for matriculation, SAT, and college exams**